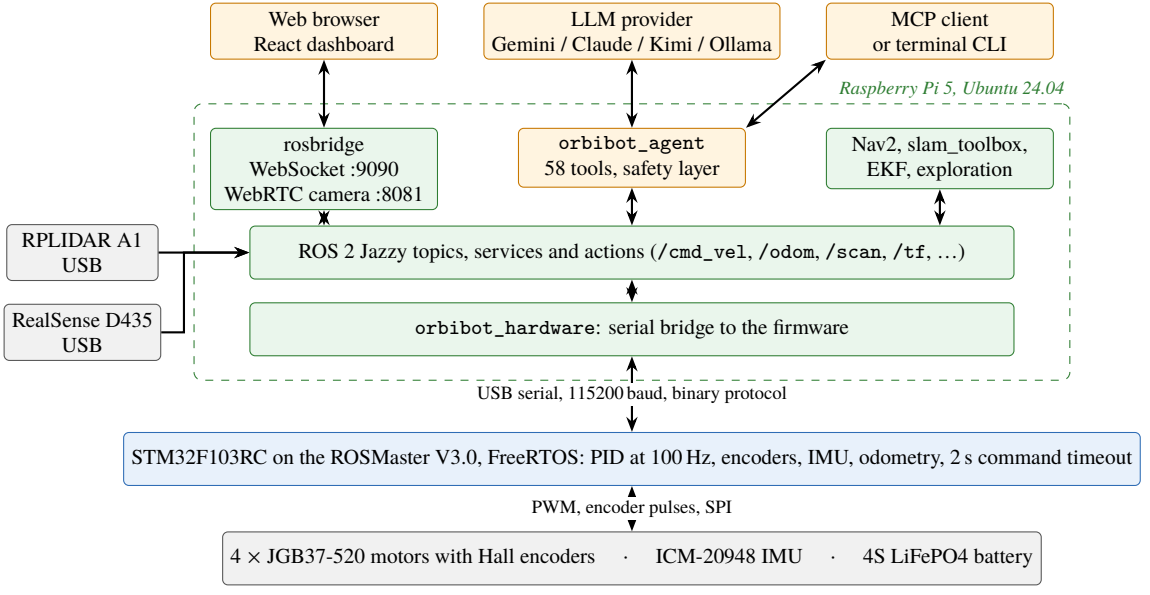
Web browser
React dashboard
LLM provider
Gemini / Claude / Kimi / Ollama
MCP client
or terminal CLI
Raspberry Pi 5, Ubuntu 24.04
rosbridge
WebSocket :9090
WebRTC camera :8081
orbibot_agent
58 tools, safety layer
Nav2, slam_toolbox,
EKF, exploration
RPLIDAR A1
USB
RealSense D435
USB
ROS 2 Jazzy topics, services and actions (/cmd_vel, /odom, /scan, /tf, …)
orbibot_hardware: serial bridge to the firmware
USB serial, 115200 baud, binary protocol
STM32F103RC on the ROSMaster V3.0, FreeRTOS: PID at 100 Hz, encoders, IMU, odometry, 2 s command timeout
PWM, encoder pulses, SPI
4 × JGB37-520 motors with Hall encoders · ICM-20948 IMU · 4S LiFePO4 battery